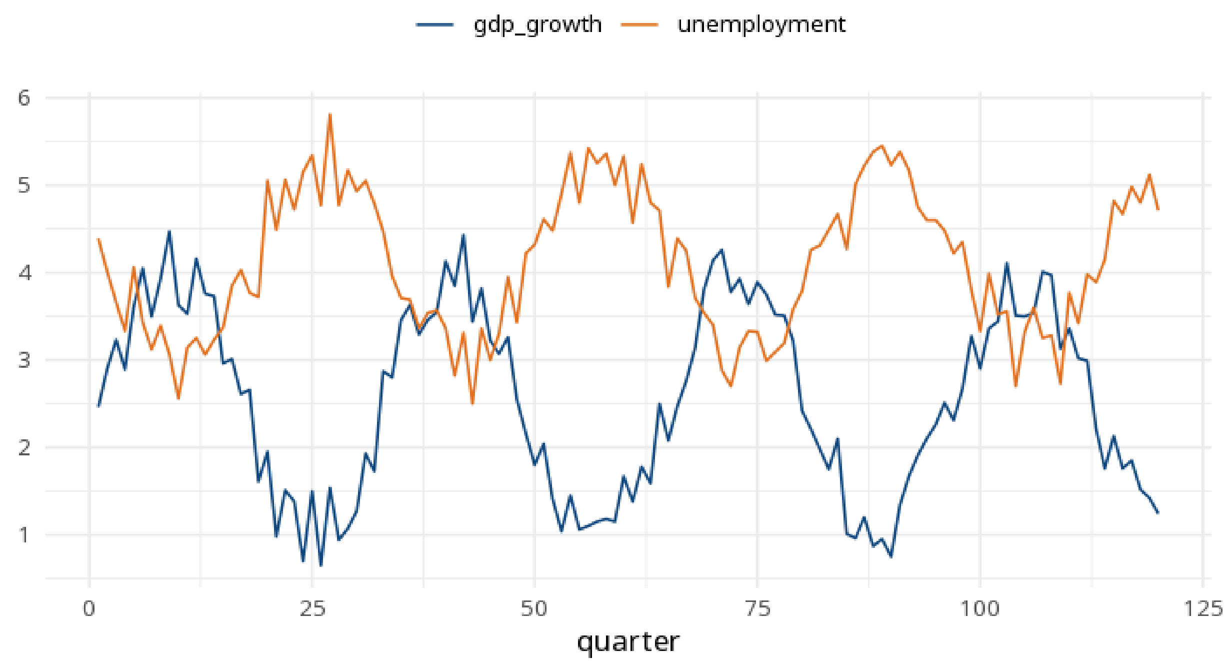
1 Granger causality

Direction	F	df	p
X→Y	122.15	4, 111	<.001
Y→X	3.42	4, 111	.011

Lag	AIC	BIC	HQ
1	−5.14	−5.00	−5.08
2	−5.14	−4.90	−5.04
3	−5.17	−4.84	−5.04
4	−5.23	−4.81	−5.06



A significant X→Y p means past values of X help predict Y beyond Y's own past — this is predictive precedence, not proof that X causes Y. Check both directions. The result depends entirely on the lag order (this card uses a max lag of 4): choose it on theory or by an information criterion — vars::VARselect reports the AIC- and BIC-minimising lag — and report it, since different lags can flip the conclusion. Make both series stationary first. Method: R lmtest::grangertest (Granger, 1969). Your significance threshold (α) is 0.05.

Granger test X→Y: $F(4,111)=122.15$, $p < .001$; Y→X: $F(4,111)=3.42$, $p = .011$ (lag=4).

Statistical basis: Granger causality F-test (Granger, C. W. J. (1969). "Investigating causal relations by econometric models and cross-spectral methods."

Econometrica, 37(3), 424-438.); Implementation (lmtest::grangertest) (Zeileis A, Hothorn T (2002). "Diagnostic Checking in Regression Relationships." R News, 2(3), 7-10.).